

Problem 9: What conditions of P and C need to exist for:

$$\vec{M}_C = \frac{d}{dt}(\vec{H}_{/C})$$

Basics:

Starting out with Newton's second law

$$\vec{F}_P = m\vec{a}_{P/O}$$

Multiply both sides by $\vec{r}_{P/C}$,

$$\vec{M}_C = \vec{r}_{P/C} \times \vec{F}_P = \vec{r}_{P/C} \times \vec{a}_{P/O}m$$

This equation is true irrespective of how C is moving in frame $\mathcal{F}$. For each of the *candidates*, we are trying to find condition of C such that $\frac{d}{dt}(\vec{H}_{/C}) = \vec{r}_{P/C} \times \vec{a}_{P/O}m$.

Candidate A: $\vec{H}_{/C} = \vec{r}_{P/C'} \times \vec{v}_{P/O} m$

where C' is a point fixed in $\mathcal{F}$ that instantaneously coincides with C.

$$\vec{H}_{/C} = \vec{r}_{P/C'} \times \vec{v}_{P/O} m$$

Taking the derivative of $\vec{H}_{/C}$ and using chain rule,

$$\frac{d}{dt}(\vec{H}_{/C}) = (\vec{v}_{P/C'} \times \vec{v}_{P/O})m + \vec{r}_{P/C'} \times \vec{a}_{P/O} m$$

Since $\vec{v}_{P/C'} = \vec{v}_{P/O} - \vec{v}_{C'/O}$, and C' is fixed in frame $\mathcal{F}$, $\vec{v}_{P/C'} = \vec{v}_{P/O}$

$$\frac{d}{dt}(\vec{H}_{/C}) = (\vec{v}_{P/C'} \times \vec{v}_{P/C'})m + \vec{r}_{P/C'} \times \vec{a}_{P/O} m$$

$$\vec{M}_C = \frac{d}{dt}(\vec{H}_{/C}) = \vec{r}_{P/C'} \times \vec{a}_{P/O} m$$

Candidate A holds true irrespective of where (location) C' is within frame $\mathcal{F}$.

Candidate B: $\vec{H}_{/C} = \vec{r}_{P/C} \times \vec{v}_{P/O} m$,

Taking the time derivative of $\vec{H}_{/C}$ and using chain rule,

$$\frac{d}{dt}(\vec{H}_{/C}) = (\vec{v}_{P/C} \times \vec{v}_{P/O})m + \vec{r}_{P/C} \times \vec{a}_{P/O}m$$

For $\vec{M}_C = \dot{\vec{H}}_{/C}$ to be consistent with Newton's 2nd Law, $(\vec{v}_{P/C} \times \vec{v}_{P/O})$ needs to be 0 since $\vec{M}_C = \dot{\vec{H}}_{/C} = \vec{r}_{P/C'} \times \vec{a}_{P/O}m$ will always hold true irrespective of how C is moving.

Expanding out the first part of equation,

$$\begin{aligned}\vec{v}_{P/C} \times \vec{v}_{P/O} &= (\vec{v}_{P/O} - \vec{v}_{C/O}) \times \vec{v}_{P/O} \\ &= -\vec{v}_{C/O} \times \vec{v}_{P/O}\end{aligned}$$

For the cases stated below $\vec{v}_{C/O} \times \vec{v}_{P/O}$ is 0, the equation above boils down to $\dot{\vec{H}}_{/C} = \vec{r}_{P/C} \times \vec{a}_{P/O}m$

For **Candidate B** to be true,

1. **C is fixed in frame $\mathcal{F}$.** Same as Candidate A
2. **C is moving at the same velocity as particle P.** $\vec{v}_{P/C} = 0$ therefore $\vec{v}_{P/C} \times \vec{v}_{P/O} = 0$.

Trivial Case:

C coincides with point P at all times (Angular momentum of a particle with respect to itself is 0)

Candidate C: $\vec{H}_{/C} = \vec{r}_{P/C} \times \vec{v}_{P/C}m,$

Taking the time derivative of $\vec{H}_{/C}$ and using chain rule,

$$\frac{d}{dt}(\vec{H}_{/C}) = (\vec{v}_{P/C} \times \vec{v}_{P/C})m + \vec{r}_{P/C} \times \vec{a}_{P/C}m$$

Expanding out $\vec{a}_{P/C} = \vec{a}_{P/O} - \vec{a}_{C/O},$

$$\begin{aligned}\frac{d}{dt}(\vec{H}_{/C}) &= \vec{r}_{P/C} \times \vec{a}_{P/C}m \\ \frac{d}{dt}(\vec{H}_{/C}) &= \vec{r}_{P/C} \times (\vec{a}_{P/O} - \vec{a}_{C/O})m\end{aligned}$$

Following the derivation in part b. the equation can be expanded to,

$$\frac{d}{dt}(\vec{H}_{/C}) = (\vec{r}_{P/C} \times m\vec{a}_{P/O}) - (\vec{r}_{P/C} \times m\vec{a}_{C/O})$$

For $\vec{M}_C = \dot{\vec{H}}_{/C} = \vec{r}_{P/C} \times m\vec{a}_{P/O},$

$(\vec{r}_{C/O} \times m\vec{a}_{C/O})$ needs to equal 0.

For **Candidate C** to be true,

1. **Point C is fixed in Frame F**
2. **Point C is moving at a constant velocity.** $\vec{a}_{C/O} = 0.$
3. $\vec{r}_{P/C}$ is parallel with $\vec{a}_{C/O}$. The relative position of P with respect to C is **parallel** to the acceleration of C with respect to O.

Trivial Case:

C coincides with point P at all times (Angular momentum of a particle with respect to itself is 0)